Business Requirement Document

Project: Pizza Sales Analysis

Business Problem

The pizza restaurant struggles to understand its sales performance and customer demand trends. Management does not have clarity on best-selling pizzas, seasonal fluctuations, customer ordering patterns or underperforming items. This limits their ability to make data-driven decisions on pricing, promotions, inventory and menu optimization.

Business Objective

The goal is to analyze historical pizza sales data to:

1. Identify top-performing and underperforming pizzas.
2. Measure total revenue and average order value.
3. Discover sales patterns across different time frames.
4. Provide actionable insights for menu optimization.
5. Support management in data-driven decision-making.

Data Source

Dataset: pizza_sales.csv

Data Description

- **order_id** → Unique identifier for each customer order
- **pizza_id** → Unique identifier for each pizza sold

- **pizza_name_id** → System-defined code representing pizza type and size
- **pizza_name** → Display name of the pizza sold
- **quantity** → Number of pizzas sold in a single order
- **unit_price** → Price of one pizza unit
- **total_price** → Total revenue generated from the transaction
- **order_date** → Date when the order was placed
- **order_time** → Time of order placement
- **pizza_size** → Size of the pizza (S, M, L, XL) for size preference analysis
- **pizza_category** → Category of pizza (Classic, Veggie, Supreme, etc.) for category-level insights
- **pizza_ingredients** → Ingredients used in the pizza

Key Performance Indicators(KPIs)

- **Total Revenue** = Sum of total_price
- **Total Pizzas Sold** = Sum of quantity
- **Total Orders** = Count of unique order_id
- **Average Order Value (AOV)** = Total Revenue ÷ Total Orders
- **Average Pizza per Order** = Total Pizzas Sold ÷ Total Orders

Business Questions

1. Which pizzas sell the most by quantity ?And which pizzas are sold the least?
2. What is the total revenue and average order value for each month?
3. On which days of the week or hours of the day are most orders placed?
4. Which pizza categories and sizes are most popular?
5. Are there seasonal trends affecting pizza sales?
6. How can promotions or menu changes increase overall revenue?
7. Which ingredients are most used?

8. Which pizza category has the highest total sales?
9. How does total revenue vary by day?
10. Which pizzas generate the most revenue? Which generate the least?
11. Which are the top 10 most profitable pizzas?
12. What are the peak ordering time for each pizza category?
13. What are the peak ordering time for each pizza size?